
Integrator Certification Brief

Aqualectra Integrator

Integrator Certification Project



CONIOG

DOCUMENT CONTROL

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Table of Contents

1. SUMMARY	4
2. ADMINISTRATION	4
3. SITE CONFIGURATION	5
3.1 Server	5
3.2 Client.....	5
3.3 Vending System.....	5
3.4 URL	5
3.5 Client Credentials	5
4. DEPLOYMENT REPORT	7
4.1 Issued Test Meter Numbers	7
4.2 Credit Vend and Reprint.....	7
4.3 Confirm Customer	8
4.4 Last Advise Response	9
4.4.1 Minimum Vend Amount	10
4.5 Issued Test Meter Numbers	10
4.6 Credit Vend and Reprint.....	10
4.7 Confirm Customer	11
4.8 Last Advise Response	12
4.8.1 Minimum Vend Amount.....	13
4.9 XML Response Electricity	13
Customer	13
Credit Vend	14
Reprint	14
Last Advice.....	15
4.10 XML Response Water	17
Customer	17
Credit Vend	18
Reprint	19
Last Advice.....	20
5. CONCLUSION	22
6. ANNEXURE	23
7. ADDITIONAL COMMENTS.....	24
8. GLOSSARY	25
9. RECERTIFICATION SIGN OFF	26

1. SUMMARY

In order to improve the transactions throughput and quality, Aqualectra Company has instructed Conlog to certify your client against a test system which replicates the live environment.

Provided in this document are the test credentials as well as meter numbers to test against. Some of these meter numbers will not work on purpose. These are designed to test the fail case scenarios of your client.

Functional Area	No. of Tests Executed	Subtotal of Tests Passed	Subtotal of Tests Failed	PassPercentage (%)	status
Credit Vend					
Reprint					
Confirm Customer					
Last Advice					
Total					

2. ADMINISTRATION

Customer	Aqualectra
Date	13/11/2018
Tester	PSBBANK NV / WebTeh
Incident No.	#INC-24755
Status	
Scope of Test	Validation Of XML request and Response Transactions: Credit Vend Reprints Confirm Customer Last Advice Response
Out of Scope	Customer Accounts Management Impulse Vending Free Issues FBE Account Payment Key Token Change

3. SITE CONFIGURATION

3.1 Server

Operating System	Windows Server 2016 Standard 64-bit
SQL Server Version	MS SQL Server 2018
Infrastructure	VPN Hosted
PowerHub Version	PowerHub 2018 Release 2 Patch 2 [6.2.0.2]
Web Browser	Internet Explorer 11 , Fire Fox 33.0.1, Google Chrome 38.0.2125.111 m
Secure Module Service Version	99.7.0.1
Language (.sib files)	-
Language	English

3.2 Client

Operating System	CentOS 7 (Application) Windows Server 2008 R2 Standard (FTP)
Infrastructure	VDC Hosted

3.3 Vending System

Ultima	UltimaPlus 2018 Release 4 [5.4.0.0]
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3.4 URL

Gateway	http://10.1.0.34:17300/GatewayServiceASP.aspx
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3.5 Client Credentials

Integrator Name	PSB Vending Unit
Client ID	VUPSB
Terminal ID	0000000000002
Operator ID	OPPSB
Password	OPPSB

4. DEPLOYMENT REPORT

In order to review your client's transactions we require that you connect your client using the above provided credentials and test URL. Please record the response from your client and take screenshots of the response. Some of these meter numbers will return errors of various natures and we would like an indication of how your client catches these. Each credit vend should be for 50 US Dollars.

4.1 Issued Test Meter Numbers Electricity

04257670465 - Active
04252652286 - Minimum Vend amount
04259063131 - Blocked customer
04262549407 - Meter not in the system / Disconnected
04004088094 - Vending server not available
04254372016 - Active

4.2 Credit Vend and Reprint

Scope of Test			Validation Of XML request and Response Transactions			
Out of Scope			Customer Accounts Management			
Test Cases						
Test Case no.	Action	Expected Result	Result			Comments
			Pass	Fail	N/A	
1	Perform a Credit Vend to the blocked meter number 04259063131	Error Message “ Purchasing is Blocked ”				
2	Perform a Credit Vend to the meter number 04257670465	Obtain Token with below details units, Proper currency symbol (P),Tariff Index, receipt Number, Customer Name, Account Number , ERF/Plot/Lot, SGC, CDU ID, Operator ID, Transaction Date, Additional Charges				
3	Perform a Reprint to the meter number blocked 04259063131	Error Message “ Purchasing is Blocked ”				
4	Perform a Reprint to the connected meter number 04257670465	Receive a response similar to Credit vend details in test case 2				
5	Perform a Credit Vend with a disconnected meter number 04262549407	Error Message “ 1216 No open account matching 04262549407 was found ”				

4.3 Confirm Customer

Scope of Test			Validation Of XML request and Response Transactions			
Out of Scope			Customer Accounts Management			
Test Cases						
Test Case no.	Action	Expected Result	Result			Comments
			Pass	Fail	N/A	
1	Perform a Confirm Customer with blocked meter number 04259063131	Error Message “ Purchasing is Blocked ”				
2	Perform a confirm Customer meter number 04257670465	Receive Customer details: Name, Address, contact, Account Number, Erf/Plot/Lot, Meter Number and <canVend>true</canVend>				
3	Perform a confirm customer with a meter that is not in the system 04004088094	Error Message “ 1216 No open account matching 04087865822 was found ”				
4	Perform a confirm customer with a meter number 04040406680 that exist on the system but assigned to a different vending server	Error Message” This client does not have access to any instances ”				Test case not required since Aqualectra only uses one vending server and it performs meter hunt for all meters not on the system.

4.4 Last Advise Response

Scope of Test			Validation Of XML request and Response Transactions			
Out of Scope			Customer Accounts Management			
Test Cases						
Test Case no.	Action	Expected Result	Result			Comments
			Pass	Fail	N/A	
1	-Perform a Credit Vend with meter number 04257670465 - Perform Last Advise Response using Date Time String and Unique Number from the Confirm Customer done above.	Receive exact response as that of the Confirm Customer on the XML response.				
2	-Perform a Credit Vend with blocked meter number 04259063131 - Perform Last Advise Response using Date Time String and Unique Number from the Confirm Customer done above.	Receive exact response as that of the Confirm Customer on the XML response.				
3	Perform Last Advise Response using above Date Time String and increment the Unique Number by 1 from the Credit Vend done above. Thus making the unique number to be invalid.	Error Message “ There was a server error obtaining the Last response”				

4.4.1 Minimum Vend Amount

Scope of Test			Validation Of XML request and Response Transactions			
Out of Scope			Customer Accounts Management			
Test Cases						
Test Case no.	Action	Expected Result	Result			Comments
			Pass	Fail	N/A	
1	Perform a Credit Vend with meter number 04252652286 for (10\$) les then 50 US Dollars	Return Error “ Credit Vend failed: PROCESS ABORTED: Amount is not sufficient to pay basic charges.				

4.5 Issued Test Meter Numbers Water

04040879803 - Active
04040408777- Minimum Vend amount
04040408744 - Blocked customer
04040408769 - Meter not in the system / Disconnected
0404088078 - Vending server not available
04040408751 - Active

4.6 Credit Vend and Reprint

Scope of Test			Validation Of XML request and Response Transactions			
Out of Scope			Customer Accounts Management			
Test Cases						
Test Case no.	Action	Expected Result	Result			Comments
			Pass	Fail	N/A	
1	Perform a Credit Vend to the blocked meter number 04040408744	Error Message “ Purchasing is Blocked ”				
2	Perform a Credit Vend to the meter number 04040879803	Obtain Token with below details units, Proper currency symbol (P),Tariff Index, receipt Number, Customer Name, Account Number , ERF/Plot/Lot, SGC, CDU ID, Operator ID, Transaction Date, Additional Charges				

3	Perform a Reprint to the meter number blocked 04040408744	Error Message “ Purchasing is Blocked ”				
4	Perform a Reprint to the connected meter number 04040879803	Receive a response similar to Credit vend details in test case 2				
5	Perform a Credit Vend with a disconnected meter number 04040408769	Error Message “ 1216 No open account matching 04040408769 was found ”				

4.7 Confirm Customer

Scope of Test			Validation Of XML request and Response Transactions			
Out of Scope			Customer Accounts Management			
Test Cases						
Test Case no.	Action	Expected Result	Result			Comments
			Pass	Fail	N/A	
1	Perform a Confirm Customer with blocked meter number 04040408744	Error Message “ Purchasing is Blocked ”				
2	Perform a confirm Customer meter number 04040879803	Receive Customer details: Name, Address, contact, Account Number, Erf/Plot/Lot, Meter Number and <canVend>true</canVend>				
3	Perform a confirm customer with a meter that is not in the system 04004088078	Error Message “ 1216 No open account matching 04004088078 was found ”				
4	Perform a confirm customer with a meter number 04040406680 that exist on the system but assigned to a different vending server	Error Message” This client does not have access to any instances ”				Test case not required since Aqualectra only uses one vending server and it performs meter hunt for all meters not on the system.

4.8 Last Advise Response

Scope of Test			Validation Of XML request and Response Transactions			
Out of Scope			Customer Accounts Management			
Test Cases						
Test Case no.	Action	Expected Result	Result			Comments
			Pass	Fail	N/A	
1	-Perform a Credit Vend with meter number 04040879803 - Perform Last Advise Response using Date Time String and Unique Number from the Confirm Customer done above.	Receive exact response as that of the Confirm Customer on the XML response.				
2	-Perform a Credit Vend with blocked meter number 04040408744 - Perform Last Advise Response using Date Time String and Unique Number from the Confirm Customer done above.	Receive exact response as that of the Confirm Customer on the XML response.				
3	Perform Last Advise Response using above Date Time String and increment the Unique Number by 1 from the Credit Vend done above. Thus making the unique number to be invalid.	Error Message “ There was a server error obtaining the Last response ”				

4.8.1 Minimum Vend Amount

Scope of Test		Validation Of XML request and Response Transactions				
Out of Scope		Customer Accounts Management				
Test Cases						
Test Case no.	Action	Expected Result	Result			Comments
			Pass	Fail	N/A	
1	Perform a Credit Vend with meter number 04252652286 for (10\$) les then 50 US Dollars	Return Error “ Credit Vend failed: PROCESS ABORTED: Amount is not sufficient to pay basic charges. ”				

XML Responses : Electricity

Confirm Customer

```
<?xml version="1.0" encoding="utf-8" ?>
- <ConfirmCustomerResp xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <clientID xsi:type="EANDeviceID" ean="VUMCB" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <serverID xsi:type="GenericDeviceID" id="1" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <terminalID xsi:type="EANDeviceID" ean="0000000000001"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <reqMsgID dateTime="20181113061935" uniqueNumber="817"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <respDateTime xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">2018-11-13T06:19:36.7788077-
    04:00</respDateTime>
  <dispHeader xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Confirm Customer</dispHeader>
  <operatorMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Confirm Customer</operatorMsg>
  <custMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Confirm Customer</custMsg>
  <utility name="Aqualectra" address="Aqualectra" taxRef="0"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
- <confirmCustResult xmlns="http://www.nrs.eskom.co.za/xmlvend/revenue/2.1/schema">
  <custVendDetail name="" address="" contactNo="" accNo="0011" locRef="Test" utilityType="Electricity"
    daysLastPurchase="5" />
- <custVendConfig>
  <canVend>true</canVend>
  <maxVendAmt value="0" symbol="ANG" />
  <minVendAmt value="0" symbol="ANG" />
  </custVendConfig>
- <meterDetail krn="1" ti="1" sgc="300601" unitOfMeasurement="kWh" msno="04040404040">
  <meterType tt="02" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  </meterDetail>
- <custAccountDetail>
  <accountStatus xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  </custAccountDetail>
  </confirmCustResult>
  </ConfirmCustomerResp>
```

Credit Vend

```
<CreditVendResp xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <clientID xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" xsi:type="EANDeviceID" ean="VUMCB" />
  <serverID xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" xsi:type="GenericDeviceID" id="1" />
  <terminalID xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" xsi:type="EANDeviceID"
ean="0000000000001" />
  <reqMsgID xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" dateTime="20181107074018"
uniqueNumber="398" />
  <respDateTime xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">2018-11-07T07:40:19.0404179-
04:00</respDateTime>
  <dispHeader xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Credit Purchase</dispHeader>
  <operatorMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Credit Purchase
Successful</operatorMsg>
  <custMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Credit Purchase</custMsg>
  <utility xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" name="Aqualectra" address="Aqualectra"
taxRef="0" />
  <clientStatus xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">
    <availCredit value="0" symbol="ANG" />
    <batchStatus banking="open" sales="open" shift="open" />
  </clientStatus>
  <vendor xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <custVendDetail xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" name=" " address=""
contactNo="" accNo="0011" locRef="Test" utilityType="Electricity" daysLastPurchase="0" />
  <creditVendReceipt xmlns="http://www.nrs.eskom.co.za/xmlvend/revenue/2.1/schema" receiptNo="48">
    <transactions>
      <tx xsi:type="CreditVendTx" receiptNo="48">
        <amt value="500" symbol="ANG" />
        <creditTokenIssue xmlns:q1="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema"
xsi:type="q1:SaleCredTokenIssue">
          <q1:desc>Credit Token</q1:desc>
          <q1:meterDetail unitOfMeasurement="kWh" msno="04040404040" sgc="300601" krn="1" ti="1">
            <q1:meterType tt="02" />
          </q1:meterDetail>
          <q1:token xsi:type="q1:STS1Token">
            <q1:stsCipher>04040404040000000020</q1:stsCipher>
          </q1:token>
          <q1:units value="2" siUnit="kWh" />
          <q1:resource xsi:type="q1:Electricity" />
        </creditTokenIssue>
        <tariff />
      </tx>
      <tx xsi:type="ServiceChrgTx">
        <amt value="0" symbol="ANG" />
        <accDesc>VAT</accDesc>
      </tx>
      <tenderAmt value="500" symbol="ANG" />
      <change value="0" symbol="ANG" />
    </transactions>
    <MonthlyTotals xsi:nil="true" />
  </creditVendReceipt>
</CreditVendResp>
```

Reprint

```
<?xml version="1.0" encoding="utf-8" ?>
- <ReprintResp xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <clientID xsi:type="EANDeviceID" ean="VUMCB" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <serverID xsi:type="GenericDeviceID" id="1" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <terminalID xsi:type="EANDeviceID" ean="0000000000001"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <reqMsgID dateTime="20181113065655" uniqueNumber="23"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <respDateTime xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">2018-11-13T06:56:57.8799456-
    04:00</respDateTime>
  <dispHeader xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Transaction Reprint</dispHeader>
  <operatorMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Transaction Reprint</operatorMsg>
  <custMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Transaction Reprint</custMsg>
  <utility name="Aqualectra" address="Aqualectra" taxRef="0"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
- <reprint xsi:type="CreditVendResp" xmlns="http://www.nrs.eskom.co.za/xmlvend/revenue/2.1/schema">
  <clientID xsi:type="EANDeviceID" ean="VUMCB" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <serverID xsi:type="GenericDeviceID" id="4" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <terminalID xsi:type="EANDeviceID" ean="0000000000001"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <reqMsgID dateTime="20181107074026" uniqueNumber="398"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <respDateTime xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">2018-11-
    07T05:40:18.231089Z</respDateTime>
  <dispHeader xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Credit Purchase</dispHeader>
  <operatorMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Credit Purchase
    Successful</operatorMsg>
  <custMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Credit Purchase</custMsg>
- <clientStatus xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">
  <availCredit value="0" symbol="ANG" />
  </clientStatus>
  <custVendDetail name="" address="" contactNo="" accNo="0011" locRef="Test" utilityType="Electricity"
    daysLastPurchase="0" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
- <creditVendReceipt receiptNo="48">
- <transactions>
- <tx xsi:type="CreditVendTx" receiptNo="48">
  <amt value="500" symbol="ANG" />
- <creditTokenIssue xmlns:q1="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema"
  xsi:type="q1:SaleCredTokenIssue">
  <q1:desc>Credit Token</q1:desc>
- <q1:meterDetail krn="1" ti="1" sgc="300601" unitOfMeasurement="kWh" msno="04040404040">
  <q1:meterType tt="02" />
  </q1:meterDetail>
- <q1:token xsi:type="q1:STS1Token">
  <q1:stsCipher>04040404040000000020</q1:stsCipher>
  </q1:token>
  <q1:units value="2" siUnit="kWh" />
  <q1:resource xsi:type="q1:Electricity" />
  </creditTokenIssue>
<tariff />
</tx>
- <tx xsi:type="ServiceChrgTx">
  <amt value="0" symbol="ANG" />
  <accDesc>VAT</accDesc>
</tx>
```

```
<tenderAmt value="500" symbol="ANG" />
<change value="0" symbol="ANG" />
  </transactions>
<MonthlyTotals xsi:nil="true" />
  </creditVendReceipt>
  </reprint>
  </ReprintResp>
```

Last Advice

```
<?xml version="1.0" encoding="utf-8" ?>
- <LastRespAdviceResp xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <clientID xsi:type="EANDeviceID" ean="VUMCB" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <serverID xsi:type="GenericDeviceID" id="1" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <terminalID xsi:type="EANDeviceID" ean="0000000000001"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <reqMsgID dateTime="20181113065927" uniqueNumber="699"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <respDateTime xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">2018-11-13T06:59:28.4304535-
    04:00</respDateTime>
  <dispHeader xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Last Response Advice</dispHeader>
  <operatorMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Last Response Advice</operatorMsg>
  <custMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <adviseReqMsgID dateTime="20181113065655" uniqueNumber="23"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
- <lastResponse xmlns:q1="http://www.nrs.eskom.co.za/xmlvend/revenue/2.1/schema" xsi:type="q1:ReprintResp"
  xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">
  <clientID xsi:type="EANDeviceID" ean="VUMCB" />
  <serverID xsi:type="GenericDeviceID" id="1" />
  <terminalID xsi:type="EANDeviceID" ean="0000000000001" />
  <reqMsgID dateTime="20181113065655" uniqueNumber="23" />
  <respDateTime>2018-11-13T06:56:57.8799456-04:00</respDateTime>
  <dispHeader>Transaction Reprint</dispHeader>
  <operatorMsg>Transaction Reprint</operatorMsg>
  <custMsg>Transaction Reprint</custMsg>
  <utility name="Aqualectra" address="Aqualectra" taxRef="0" />
- <q1:reprint xsi:type="q1:CreditVendResp">
  <clientID xsi:type="EANDeviceID" ean="VUMCB" />
  <serverID xsi:type="GenericDeviceID" id="4" />
  <terminalID xsi:type="EANDeviceID" ean="0000000000001" />
  <reqMsgID dateTime="20181107074026" uniqueNumber="398" />
  <respDateTime>2018-11-07T05:40:18.231089Z</respDateTime>
  <dispHeader>Credit Purchase</dispHeader>
  <operatorMsg>Credit Purchase Successful</operatorMsg>
  <custMsg>Credit Purchase</custMsg>
- <clientStatus>
  <availCredit value="0" symbol="ANG" />
  </clientStatus>
  <custVendDetail name="" address="" contactNo="" accNo="0011" locRef="Test" utilityType="Electricity"
    daysLastPurchase="0" />
- <q1:creditVendReceipt receiptNo="48">
- <q1:transactions>
- <q1:tx xsi:type="q1:CreditVendTx" receiptNo="48">
  <q1:amt value="500" symbol="ANG" />
- <q1:creditTokenIssue xsi:type="SaleCredTokenIssue">
  <desc>Credit Token</desc>
```

```
- <meterDetail krn="1" ti="1" sgc="300601" unitOfMeasurement="kWh" msno="04040404040">
  <meterType tt="02" />
</meterDetail>
- <token xsi:type="STS1Token">
  <stsCipher>04040404040000000020</stsCipher>
</token>
<units value="2" siUnit="kWh" />
<resource xsi:type="Electricity" />
</q1:creditTokenIssue>
<q1:tariff />
</q1:tx>
- <q1:tx xsi:type="q1:ServiceChrgTx">
  <q1:amt value="0" symbol="ANG" />
  <q1:accDesc>VAT</q1:accDesc>
</q1:tx>
<q1:tenderAmt value="500" symbol="ANG" />
<q1:change value="0" symbol="ANG" />
</q1:transactions>
<q1:MonthlyTotals xsi:nil="true" />
</q1:creditVendReceipt>
</q1:reprint>
</lastResponse>
</LastRespAdviceResp>
```

XML Responses : Water

Confirm Customer

```
<?xml version="1.0" encoding="utf-8" ?>
- <ConfirmCustomerResp xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <clientID xsi:type="EANDeviceID" ean="VUMCB" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <serverID xsi:type="GenericDeviceID" id="1" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <terminalID xsi:type="EANDeviceID" ean="0000000000001"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <reqMsgID dateTime="20181113093050" uniqueNumber="407"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <respDateTime xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">2018-11-13T09:30:50.9164922-
    04:00</respDateTime>
  <dispHeader xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Confirm Customer</dispHeader>
  <operatorMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Confirm Customer</operatorMsg>
  <custMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Confirm Customer</custMsg>
  <utility name="Aqualectra" address="Aqualectra" taxRef="0"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
- <confirmCustResult xmlns="http://www.nrs.eskom.co.za/xmlvend/revenue/2.1/schema">
  <custVendDetail name="" address="" contactNo="" accNo="1001" locRef="0104093" utilityType="Water"
    daysLastPurchase="0" />
- <custVendConfig>
  <canVend>true</canVend>
  <maxVendAmt value="0" symbol="ANG" />
  <minVendAmt value="0" symbol="ANG" />
</custVendConfig>
- <meterDetail krn="1" ti="1" sgc="300601" unitOfMeasurement="Kiloliters (kl)" msno="04040879803">
  <meterType tt="02" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
</meterDetail>
- <custAccountDetail>
  <accountStatus xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
```



```
</custAccountDetail>  
</confirmCustResult>  
</ConfirmCustomerResp>
```

Credit Vend

```
<?xml version="1.0" encoding="utf-8" ?>  
- <CreditVendResp xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">  
  <clientID xsi:type="EANDeviceID" ean="VUMCB" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />  
  <serverID xsi:type="GenericDeviceID" id="1" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />  
  <terminalID xsi:type="EANDeviceID" ean="0000000000001"  
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />  
  <reqMsgID dateTime="20181113093234" uniqueNumber="827"  
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />  
  <respDateTime xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">2018-11-13T09:32:35.7871862-  
    04:00</respDateTime>  
  <dispHeader xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Credit Purchase</dispHeader>  
  <operatorMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Credit Purchase  
    Successful</operatorMsg>  
  <custMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Credit Purchase</custMsg>  
  <utility name="Aqualectra" address="Aqualectra" taxRef="0"  
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />  
- <clientStatus xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">  
  <availCredit value="0" symbol="ANG" />  
  <batchStatus banking="open" sales="open" shift="open" />  
  </clientStatus>  
  <vendor xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />  
  <custVendDetail name="" address="" contactNo="" accNo="1001" locRef="0104093" utilityType="Water"  
    daysLastPurchase="0" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />  
- <creditVendReceipt receiptNo="108" xmlns="http://www.nrs.eskom.co.za/xmlvend/revenue/2.1/schema">  
- <transactions>  
- <tx xsi:type="CreditVendTx" receiptNo="108">  
  <amt value="1000" symbol="ANG" />  
- <creditTokenIssue xmlns:q1="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema"  
  xsi:type="q1:SaleCredTokenIssue">  
  <q1:desc>Credit Token</q1:desc>  
- <q1:meterDetail krn="1" ti="1" sgc="300601" unitOfMeasurement="Kiloliters (kl)" msno="04040879803">  
  <q1:meterType tt="02" />  
  </q1:meterDetail>  
- <q1:token xsi:type="q1:STS1Token">  
  <q1:stsCipher>04040879803000000040</q1:stsCipher>  
  </q1:token>  
  <q1:units value="4" siUnit="Kiloliters (kl)" />  
  <q1:resource xsi:type="q1:Water" />  
  </creditTokenIssue>  
  <tariff />  
  </tx>  
- <tx xsi:type="ServiceChrgTx">  
  <amt value="0" symbol="ANG" />  
  <accDesc>VAT</accDesc>  
  </tx>  
  <tenderAmt value="1000" symbol="ANG" />  
  <change value="0" symbol="ANG" />  
  </transactions>  
  <MonthlyTotals xsi:nil="true" />  
  </creditVendReceipt>
```

</CreditVendResp>

Reprint

```
<?xml version="1.0" encoding="utf-8" ?>
- <ReprintResp xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <clientID xsi:type="EANDeviceID" ean="VUMCB" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <serverID xsi:type="GenericDeviceID" id="1" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <terminalID xsi:type="EANDeviceID" ean="0000000000001"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <reqMsgID dateTime="20181113093331" uniqueNumber="17"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <respDateTime xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">2018-11-13T09:33:32.7838739-
    04:00</respDateTime>
  <dispHeader xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Transaction Reprint</dispHeader>
  <operatorMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Transaction Reprint</operatorMsg>
  <custMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Transaction Reprint</custMsg>
  <utility name="Aqualectra" address="Aqualectra" taxRef="0"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
- <reprint xsi:type="CreditVendResp" xmlns="http://www.nrs.eskom.co.za/xmlvend/revenue/2.1/schema">
  <clientID xsi:type="EANDeviceID" ean="VUMCB" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <serverID xsi:type="GenericDeviceID" id="4" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <terminalID xsi:type="EANDeviceID" ean="0000000000001"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <reqMsgID dateTime="20181113093243" uniqueNumber="827"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <respDateTime xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">2018-11-
    13T07:32:36.0344761Z</respDateTime>
  <dispHeader xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Credit Purchase</dispHeader>
  <operatorMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Credit Purchase
    Successful</operatorMsg>
  <custMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Credit Purchase</custMsg>
- <clientStatus xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">
  <availCredit value="0" symbol="ANG" />
  </clientStatus>
  <custVendDetail name="" address="" contactNo="" accNo="1001" locRef="0104093" utilityType="Water"
    daysLastPurchase="0" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
- <creditVendReceipt receiptNo="108">
- <transactions>
- <tx xsi:type="CreditVendTx" receiptNo="108">
  <amt value="1000" symbol="ANG" />
- <creditTokenIssue xmlns:q1="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema"
  xsi:type="q1:SaleCredTokenIssue">
  <q1:desc>Credit Token</q1:desc>
- <q1:meterDetail krn="1" ti="1" sgc="300601" unitOfMeasurement="Kiloliters (kl)" msno="04040879803">
  <q1:meterType tt="02" />
  </q1:meterDetail>
- <q1:token xsi:type="q1:STS1Token">
  <q1:stsCipher>04040879803000000040</q1:stsCipher>
  </q1:token>
  <q1:units value="4" siUnit="Kiloliters (kl)" />
  <q1:resource xsi:type="q1:Water" />
</creditTokenIssue>
```

```
<tariff />
</tx>
- <tx xsi:type="ServiceChrgTx">
  <amt value="0" symbol="ANG" />
  <accDesc>VAT</accDesc>
  </tx>
  <tenderAmt value="1000" symbol="ANG" />
  <change value="0" symbol="ANG" />
  </transactions>
  <MonthlyTotals xsi:nil="true" />
  </creditVendReceipt>
  </reprint>
  </ReprintResp>
```

Last Advice

```
<?xml version="1.0" encoding="utf-8" ?>
- <LastRespAdviceResp xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <clientID xsi:type="EANDeviceID" ean="VUMCB" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <serverID xsi:type="GenericDeviceID" id="1" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <terminalID xsi:type="EANDeviceID" ean="0000000000001"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <reqMsgID dateTime="20181113093611" uniqueNumber="396"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <respDateTime xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">2018-11-13T09:36:12.1490668-
    04:00</respDateTime>
  <dispHeader xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Last Response Advice</dispHeader>
  <operatorMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">Last Response Advice</operatorMsg>
  <custMsg xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
  <adviceReqMsgID dateTime="20181113093050" uniqueNumber="407"
    xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema" />
- <lastResponse xmlns:q1="http://www.nrs.eskom.co.za/xmlvend/revenue/2.1/schema"
  xsi:type="q1:ConfirmCustomerResp" xmlns="http://www.nrs.eskom.co.za/xmlvend/base/2.1/schema">
  <clientID xsi:type="EANDeviceID" ean="VUMCB" />
  <serverID xsi:type="GenericDeviceID" id="1" />
  <terminalID xsi:type="EANDeviceID" ean="0000000000001" />
  <reqMsgID dateTime="20181113093050" uniqueNumber="407" />
  <respDateTime>2018-11-13T09:30:50.9164922-04:00</respDateTime>
  <dispHeader>Confirm Customer</dispHeader>
  <operatorMsg>Confirm Customer</operatorMsg>
  <custMsg>Confirm Customer</custMsg>
  <utility name="Aqualectra" address="Aqualectra" taxRef="0" />
- <q1:confirmCustResult>
  <q1:custVendDetail name="" address="" contactNo="" accNo="1001" locRef="0104093" utilityType="Water"
    daysLastPurchase="0" />
- <q1:custVendConfig>
  <q1:canVend>true</q1:canVend>
  <q1:maxVendAmt value="0" symbol="ANG" />
  <q1:minVendAmt value="0" symbol="ANG" />
  </q1:custVendConfig>
- <q1:meterDetail krn="1" ti="1" sgc="300601" unitOfMeasurement="Kiloliters (kl)" msno="04040879803">
  <meterType tt="02" />
  </q1:meterDetail>
- <q1:custAccountDetail>
  <accountStatus />
  </q1:custAccountDetail>
```

</q1:confirmCustResult>
</lastResponse>
</LastRespAdviceResp>

5. CONCLUSION

6. ANNEXURE

7. ADDITIONAL COMMENTS

8. GLOSSARY

Abbreviation	Definition
XML	Extreme Markup Language
FBE	Free Basic Electricity

9. RECERTIFICATION SIGN OFF

..... Services Technician Signature	Name: Kerwin Sanssoucie
Date accepted: YYYY/MM/DD	
..... Project Engineer Signature	Name: Kevin Chetty
Date accepted: YYYY/MM/DD	
..... Applications Engineering Specialist Signature	Name: Matthew Jackson
Date accepted: YYYY/MM/DD	
..... Project & Services Manager Signature	Name: Martin Keyter
Date accepted: YYYY/MM/DD	
..... Integrator Validation	Name: (CUSTOMER NAME)
Date accepted: 2016/02/	